Higher National Diploma in Civil Engineering

Fourth Year, First Semester Examination – 2018

CE3225- Buildings and Structures

Instructions for Candidates:	No. of questions	: 06
Answer any **Five (05)** questions.	No. of pages	: 02
All question carry equal marks	Time	: Three (03) hours

1.

I. Explain the desk study and reconnaissance survey in site investigation. (08 Marks)

II. State the UDA regulations for stairs in public building with neat sketch considering following components. (06 Marks)

a) Width (W)

b) Minimum head room

c) Riser (R) and tread(T)

III. Explain why Environmental Impact Assessment required for high rise building project. (06 Marks)

2.

I. Briefly explain the functional component of solid waste management (04 Marks)

II. Briefly explain the hazardous waste (04 Marks)

III. Explain the following process in waste water treatment

a) Flocculation

b) Sedimentation

c) Disinfection (06 Marks)

IV. Discuss the advantages and limitations of up feed and down feed water supply system in a multi-story building (06 Marks)

3.

I. Explain the importance of daylighting in building design. (08 Marks)

II. Define the following terms used in lighting design.

a) Lumen-lm

b) Lux-lx

c) Illuminance

d) Glare (08 Marks)

III. Briefly explain the design consideration of lighting system of a building. (04 Marks)

4.

I. Explain the active and passive techniques used in space cooling and heating in a building (06 Marks)

II. Classify the types of fire. (04 Marks)

III. Briefly explain the protection of the structural elements of the building from fire. (04 Marks)

IV. Explain the types of firefighting system in a multi-story building. (06 Marks)

5.

I. Briefly explain the circulation of building. (04 Marks)

II. Describe the situation where you would recommend a ramp and an escalator. (06 Marks)

III. Explain the structural consideration taken in to account for lift installation. (06 Marks)

IV. Briefly explain the necessity of Rope access in multi-story building construction. (04 Marks)

6.

I. Describe the concept of sustainable building design. (05 Marks)

II. Briefly explain concept of green building. (05 Marks)

III. Discuss the advantages of green building over conventional building. (05 Marks)

IV. List five requirements of the LEED certification. (05 Marks)